

Exercise class 4

1. Properties of the Fourier transform:
 - a) Show that if $f(x)$ is a real function, then $\hat{f}(\omega)^* = \hat{f}(-\omega)$.
 - b) Show that $g(x) = f(-x)$ implies $\hat{g}(\omega) = \hat{f}(-\omega)$.
Example: Calculate the Fourier transform of $e^{-|x|}$.
 - c) Show that the Fourier transform of $g(x) = f(ax)$ is $\hat{g}(\omega) = \frac{1}{a} \hat{f}\left(\frac{\omega}{a}\right)$.
 - d) Show that the Fourier transform of $g(x) = f(x + a)$ is $\hat{g}(\omega) = e^{i\omega a} \hat{f}(\omega)$.
 - e) Show that the Fourier transform of $g(x) = f'(x)$ is $\hat{g}(\omega) = i\omega \hat{f}(\omega)$.
2. Calculation of Fourier transforms
 - a. $f(x) = e^{-a x^2}$
 - b. $f(x) = e^{-a (x-b)^2}$
 - c. $f(x) = e^{-a x^2 + i b x}$
 - d. $f(x) = x^n e^{-a x^2}$
 - e. $f(x) = x^n e^{-a |x|}$
3. Fourier transform the functions below:
 - a. $f(t) = e^{i\omega_0 t}$
 - b. $f(t) = \cos(\omega_0 t)$
 - c. $f(t) = \cos(\omega_1 t) \cos(\omega_2 t)$
4. Prove that using both the Fourier transforms above, and directly using the exponential form of the cosine.
$$\cos(\omega_1 t) \cos(\omega_2 t) = \frac{1}{2} (\cos((\omega_1 + \omega_2)t) + \cos((\omega_1 - \omega_2)t))$$
5. Low pass filter, AM radio receiver (see notes).